

Matrix discrepancy and optimization

The Beck–Fiala discrepancy conjecture

Difficulty: extreme **Importance:** broadly interesting**Status:** Partially resolved**Provenance:** source-stated conjecture.**Rating rationale:** The unrestricted square-root sparsity law is a central discrepancy conjecture with broad consequences for combinatorics, rounding and optimization.Does a finite universal constant $C > 0$ exist such that the following holds for every pair of positive integers m, n , every integer t with $1 \leq t \leq m$, and every matrix $A = (a_{ij}) \in \{0, 1\}^{m \times n}$? If

$$\sum_{i=1}^m a_{ij} \leq t \quad (1 \leq j \leq n),$$

then some signing $x \in \{-1, 1\}^n$ satisfies

$$\|Ax\|_\infty \leq C\sqrt{t}.$$

The constant must be independent of m, n, t . All columns are available when choosing the signing. The zero-sparsity case is trivial and excluded from the quantifiers.The matrix is the incidence matrix of a set system: each element belongs to at most t sets, and the signing should balance every set. The problem asks whether column sparsity alone controls the simultaneous row error at its conjectured square-root scale. It is a special case of the Komlós conjecture after column normalization, but an independent classical source-stated conjecture for incidence matrices.

References

1. D. J. Altschuler and K. Tikhomirov, *Online Beck–Fiala Down to Logarithmic Sparsity*, arXiv:2607.14238 (2026), abstract and introduction: the unrestricted conjecture and the proved sparsity regime. [Paper](#).
2. N. Bansal and H. Jiang, *Decoupling via Affine Spectral-Independence: Beck-Fiala and Komlós Bounds Beyond Banaszczyk*, arXiv:2508.03961v2 (2025), abstract and introductory results. [Paper](#).
3. J. Beck and T. Fiala, “Integer-making” theorems, *Discrete Applied Mathematics* 3(1) (1981), pp. 1–8, the original bounded-column-sparsity discrepancy results. [Article](#).

Status check — 2026-09-10

checked 2607.14238v1 (2026-07-15), 2508.03961v2 (2025-09-09), and searches “Beck Fiala conjecture solved 2026 logarithmic sparsity” and “Online Beck Fiala Down to Logarithmic Sparsity”. The 2026 work reaches sparsities $t \geq (\log n)^{1+o(1)}$ in its notation translated to n columns; it does not establish the assertion for all sparsities. Online lower bounds concern a more restrictive information model. No full resolution or withdrawal was found.**Audit update (2026-09-10):** Rechecked the July 2026 online/offline paper and searched for unrestricted Beck–Fiala results. It proves the offline target in a substantial sparsity regime, which is part of the displayed family, but leaves smaller sparsities unresolved. This is a bounded literature check, not a proof that no solution exists.